

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-405 — Android Programming

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

SECTION — A

Long-Answer Questions. Answer any two questions:

2 × 12 = 24 Marks

Question 1. Discuss the Android Architecture. Explain the importance and functions of the different layers in detail.

Question 2. Write a Flutter code snippet to create a simple application with a single page containing a centered “Hello, Flutter!” text. Also explain the concept of state management in Flutter and discuss the different approaches, providing examples of when to use each.

Question 3. Write the Flutter code to build an application with the following specifications:

- (a) The first page of the application should have an AppBar with the text “Flutter App”.
- (b) The first page should have a body with the text “First Page” and an image from the assets folder named “flutter.jpg”.
- (c) The application should have a floating action button at the bottom of the screen which navigates to the second page.
- (d) The second page should contain an AppBar with the text “Second Page” and a back button that returns to the homepage.
- (e) The second page should have a body with the text “Second Page” and an image from the assets folder named “second.jpg”.

Question 4. What is ART (Android Run Time)? Discuss the features and inner working of ART in detail.

SECTION — B

Question 5. Attempt any six parts (Answer in about 200 words each): **6 × 6 = 36 Marks**

- (a) What is the `AndroidManifest.xml` file? Write down its purpose and list its key components.
- (b) What is an APK file? Discuss in detail how you can build an APK file in Flutter.
- (c) Discuss the directory structure of a standard Android project.

- (d) What is the use of the `ListView.builder` widget in Flutter? Explain its different components using an example.
- (e) What are the key differences between Dalvik Virtual Machine (DVM) and Android Runtime (ART)?
- (f) What is Flutter? Write down some of the main disadvantages of using the Flutter framework.
- (g) What is the purpose of the `pubspec.yaml` file in a Flutter project? Explain the process of adding external dependencies (packages) using this file with an example.
- (h) Discuss the importance of using `async` and `await` in Flutter when working with asynchronous operations. Provide a code example.
- (i) Write a Flutter Dart code snippet to create a simple app that displays a list of items with images and text. Include the widget tree, data source, and any relevant packages.

SECTION — C

Question 6. Attempt all parts:

10 × 1 = 10 Marks

- (a) Which programming language is primarily used for Flutter app development?
- (b) Which programming language is primarily used for native Android app development?
- (c) Which Flutter command is used to create a new project?
- (d) Which Flutter widget is commonly used for creating scrollable lists of items?
- (e) Which file format is commonly used for storing persistent relational database data in Android?
- (f) What is the purpose of the `setState` method in Flutter?
- (g) What is the primary advantage of using Flutter for cross-platform app development?
- (h) Which Flutter widget is used for navigation between different screens or pages in an app?
- (i) What is the primary role of the `main.dart` file in a Flutter project?
- (j) What was the name of the first commercial Android smartphone?